



Functions and Algebra 3i

Subject & topics: Maths Stage & difficulty: A Level P2

The functions f and g are defined for all real values of x by

$$f(x) = |2x + a| + 3a \quad \text{and} \quad g(x) = 5x - 4a,$$

where a is a positive constant.

Part A

Range

Find the range of $f(x)$.

Fill in the inequality below.

Items:

$f(x)$

$<$

$\leq$

$>$

$\geq$

$< f(x) <$

$\leq f(x) \leq$

$< f(x) \text{ or } f(x) <$

$\leq f(x) \text{ or } f(x) \leq$

$\frac{a}{3}$

$\frac{a}{2}$

a

$2a$

$3a$

$4a$

0

$-\frac{a}{3}$

$-\frac{a}{2}$

$-a$

$-2a$

Part B

Inverse function of $f(x)$

Fill in the blanks to explain why the function $f(x)$ has no inverse.

The function $f(x)$ is not . For example, $f(0) = 4a$ and $f(\text{$) also equals $4a$. Hence, $f(x)$ has no inverse.

Items:

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-
-
-

Part C

Inverse function of $g(x)$

Find an expression for $g^{-1}(x)$.

The following symbols may be useful: a, x

Part D

Solve for x

Solve for x the equation $g(f(x)) = 31a$. Give your solutions in ascending order.

- $x = \text{} a$
- $x = \text{} a$

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Inverse Quadratic Function

Subject & topics: Maths | Functions | Graph Sketching **Stage & difficulty:** A Level P2

Figure 1 shows the graph of $y = f(x)$, where

$$f(x) = 2 - x^2, \quad x \leq 0$$

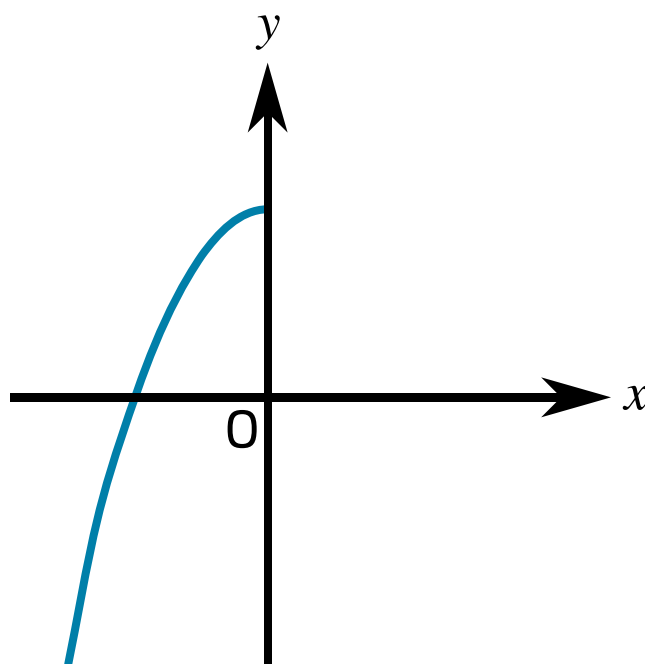


Figure 1: The graph of $y = f(x)$, for $x \leq 0$.

Part A

$$f^2(-3)$$

Evaluate $f^2(-3)$.

Part B

$f^{-1}(x)$

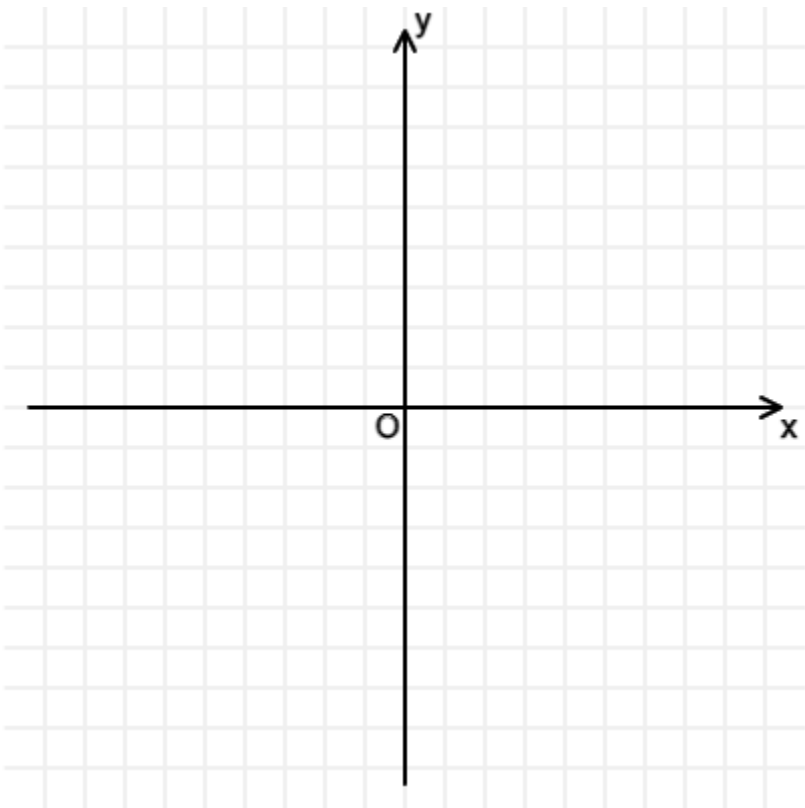
Find an expression for $f^{-1}(x)$.

The following symbols may be useful: f , x , y

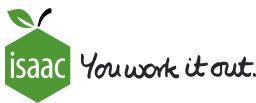
Part C

Graph of $f^{-1}(x)$

Sketch the graph of $y = f^{-1}(x)$.



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Function Types and Inverses

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level C1

Figure 1 shows five different graphs, A, B, C, D and E, each for values of x such that $-a \leq x \leq a$ where a is a constant.

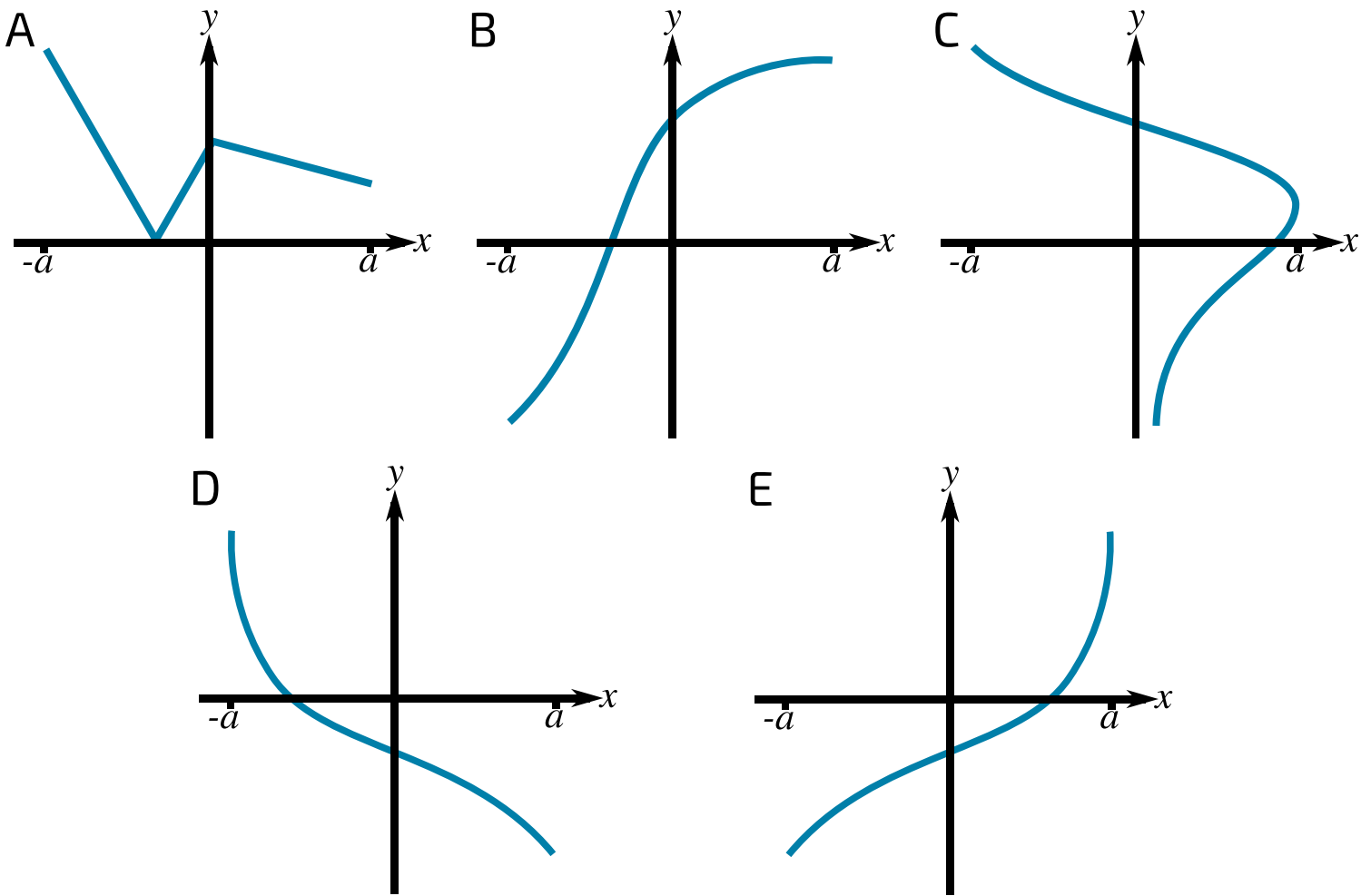


Figure 1: The set of five graphs, labelled A, B, C, D and E

Part A

Function

Which diagram does not show the graph of a function?

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

Part B

One-to-one Function

Which diagram shows the graph of a function that is not one-to-one?

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

Part C

Inverses

It is given that two of the diagrams illustrate functions that are inverses of each other. Identify one of these two diagrams.

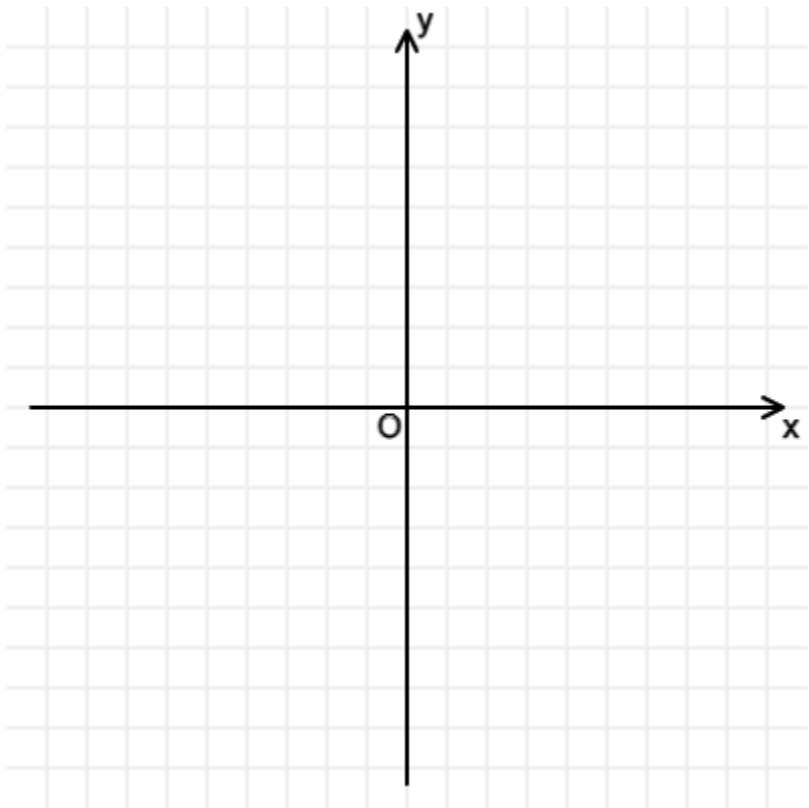
- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

Part D

Sketch

The graph in E has equation $y = f(x)$. Sketch the graph of $y = |f(x)|$.

To prevent any sharp changes in your curve from being smoothed out, sketch your curve as two sections.



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Combined Transformations

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

The function f is defined by $f(x) = \sqrt{mx + 7} - 4$, where $x \geq -\frac{7}{m}$ and m is a positive constant. **Figure 1** shows the curve $y = f(x)$.

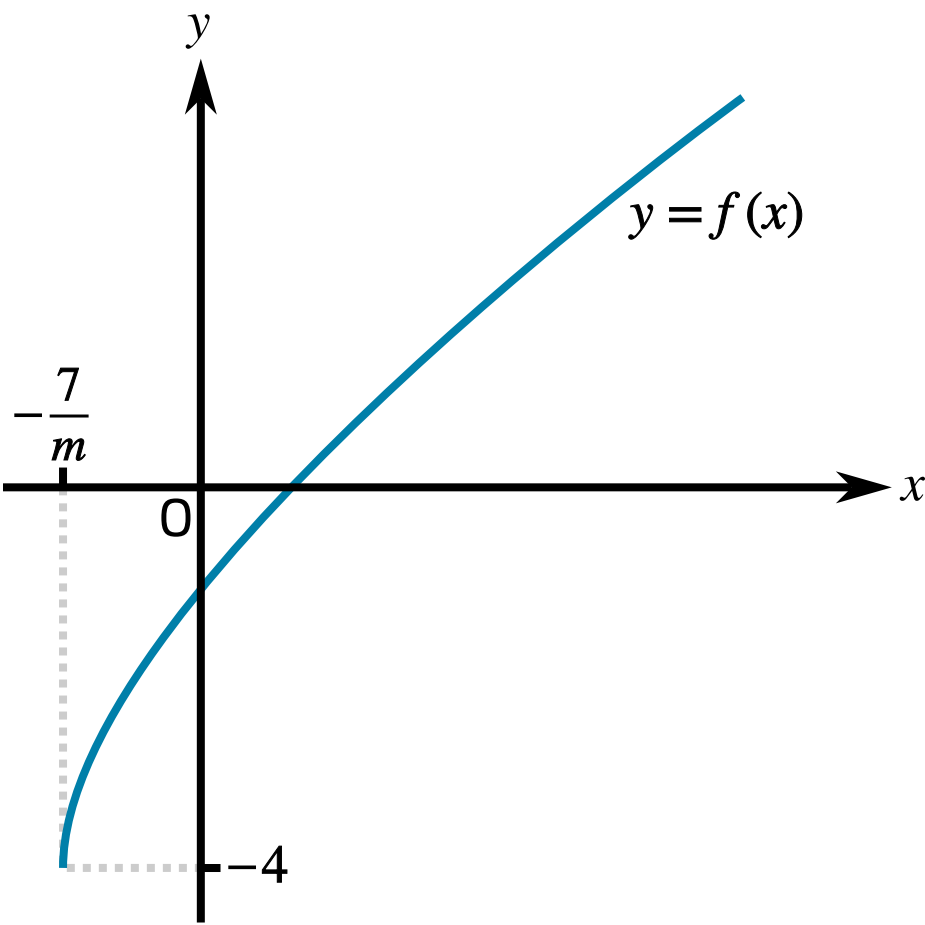


Figure 1: The curve $y = f(x)$

Part A

Translation of the curve $y = \sqrt{x}$

A sequence of transformations maps the curve $y = \sqrt{x}$ to the curve $y = f(x)$. Give details of these transformations.

Available items

Translate the curve 4 units in the negative x direction.

Translate the curve 7 units in the negative x direction.

Translate the curve 7 units in the positive x direction.

Translate the curve 4 units in the negative y direction.

Translate the curve 7 units in the negative y direction.

Stretch the curve in the y direction by a factor of $\frac{1}{m}$.

Translate the curve 4 units in the positive y direction.

Stretch the curve in the x direction by a factor of m .

Stretch the curve in the x direction by a factor of $\frac{1}{m}$.

Part B

$f^{-1}(x)$

Find an expression for $f^{-1}(x)$.

The following symbols may be useful: f , m , x

Part C

Values of m

It is given that the curves $y = f(x)$ and $y = f^{-1}(x)$ do not meet. Thus it can be deduced that neither curve meets the line $y = x$. Hence determine the possible values of m .

Construct your answer from the items below.

Items:

m

$<$

$\leq$

$>$

$\geq$

$< m <$

$\leq m \leq$

$> m \text{ or } m >$

$\geq m \text{ or } m \geq$

-28

-14

-8

-7

-4

-2

-1

0

1

2

4

7

8

14

28

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Question deck:

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Curve Sketching and Combined Transformations 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

The function f is defined for all real values of x by

$$f(x) = k(x^2 + 4x)$$

where k is a positive constant. **Figure 1** shows the curve with equation $y = f(x)$.

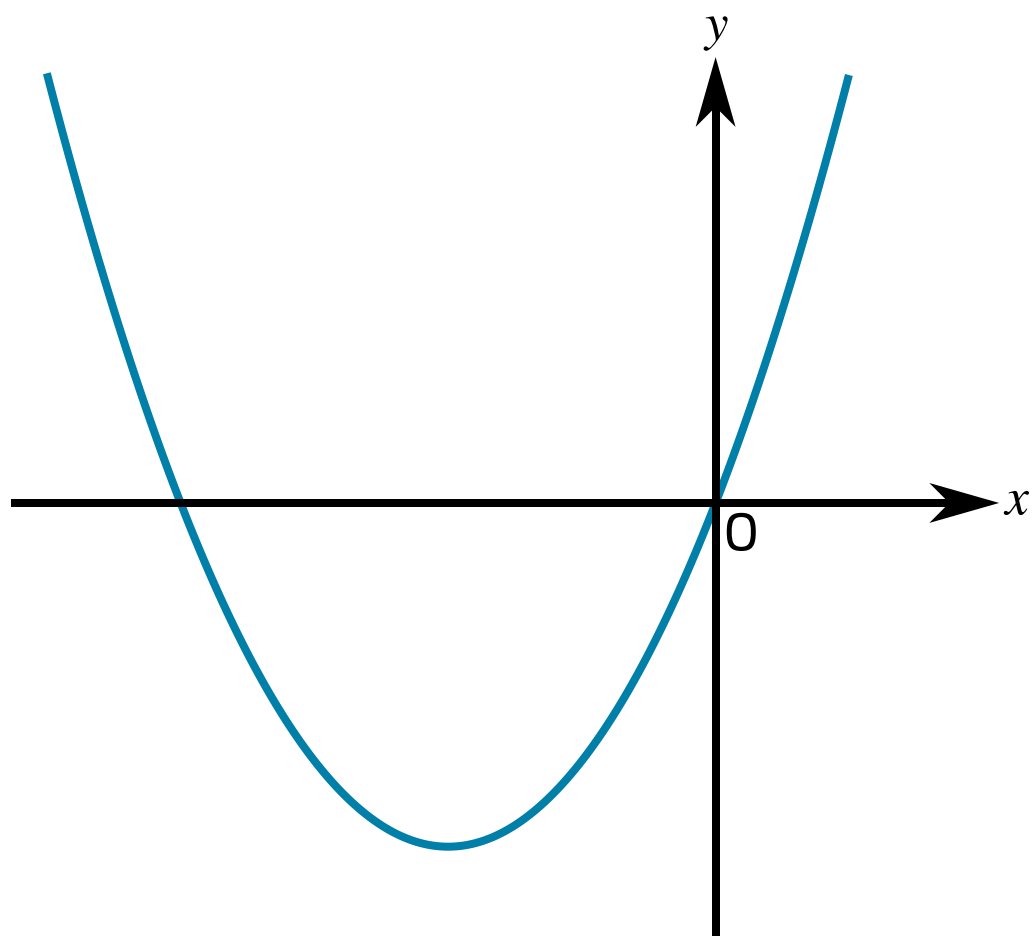


Figure 1: The graph of $y = f(x)$

Part A

Transformations

Give a sequence of transformations that will transform the curve $y = x^2$ to the curve $y = f(x)$.

Available items

- Translate the curve 2 units in the negative x direction.
- Translate the curve 4 units in the negative x direction.
- Translate the curve 4 units in the negative y direction.
- Stretch the curve in the x direction by a factor of $\frac{1}{k}$.
- Stretch the curve in the y direction by a factor of k .
- Translate the curve 4 units in the positive x direction.
- Translate the curve 4 units in the positive y direction.
- Stretch the curve in the y direction by a factor of $\frac{1}{k}$.
- Translate the curve 2 units in the positive x direction.
- Translate the curve 2 units in the positive y direction.
- Stretch the curve in the x direction by a factor of k .
- Translate the curve 2 units in the negative y direction.

Part B

Range

Find the range of $f(x)$ as a single inequality in terms of k .

The following symbols may be useful: $<$, $<=$, $>$, $>=$, $f(x)$, k , x , y

Part C

Find k

It is given that there are three distinct values of x which satisfy the equation $|f(x)| = 20$.

Find the value of k .

Part D

Solve $|f(x)| = 20$

Using the value of k from part C, find the three distinct values of x which satisfy the equation $|f(x)| = 20$.
Give any irrational values to 3 sf.

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Modulus Functions 2

Pre-Uni Maths for Science E4.6

Subject & topics: Maths | Functions | General Functions Stage & difficulty: A Level P2

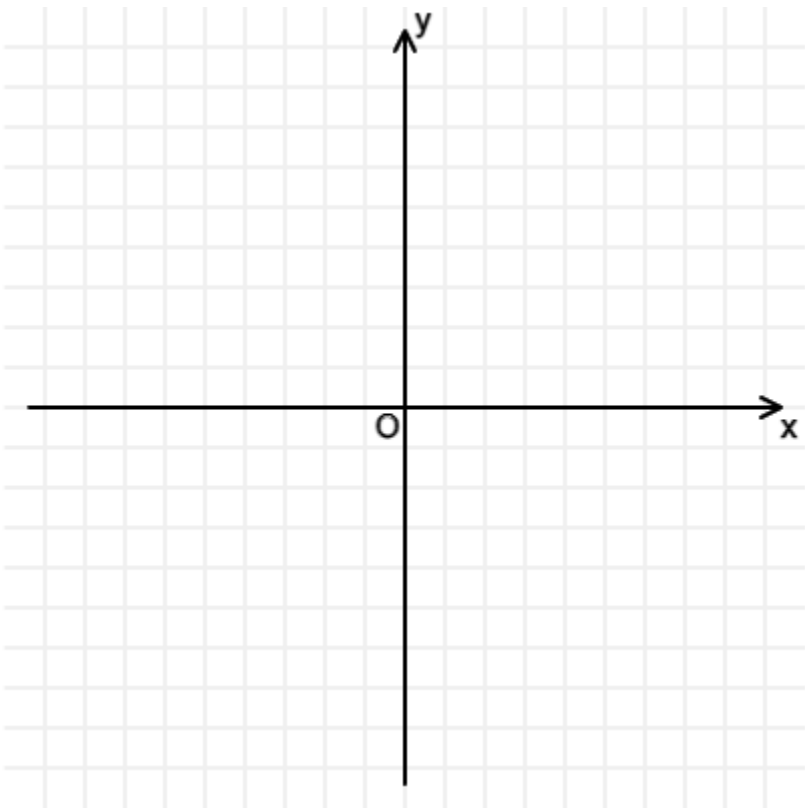
Part A

$y = \left| \frac{1}{x} \right|$

State the value of x where the function $y = \left| \frac{1}{x} \right|$ diverges.

Sketch the graph of $y = \left| \frac{1}{x} \right|$.

You do not need to draw the asymptotes.



Part B

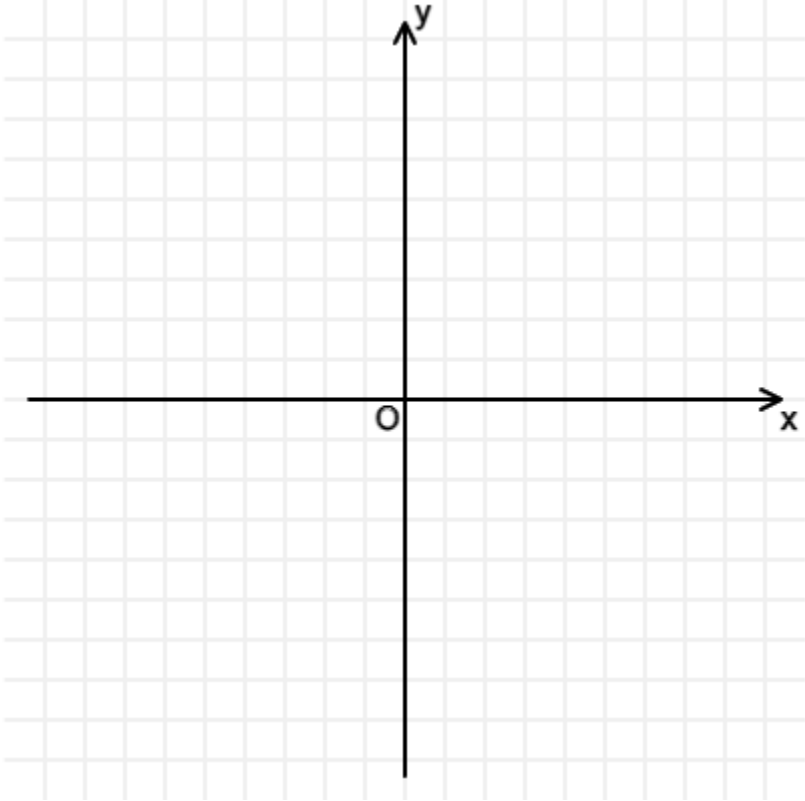
$y = \left| \frac{1}{x^2 - 4} \right|$

State the values of x where the function $y = \left| \frac{1}{x^2 - 4} \right|$ diverges.

The following symbols may be useful: $x, \pm$

Sketch the graph of $y = \left| \frac{1}{x^2 - 4} \right|$.

You do not need to draw the asymptotes.



Part C

Solve equation graphically

Solve the equation $|x| = \left| \frac{1}{x} \right|$ graphically and give the solution as a single expression.

The following symbols may be useful: $x, \pm$



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Sketching a Cubic Modulus Function

Pre-Uni Maths for Sciences E4.7

Subject & topics: Maths | Functions | Graph Sketching Stage & difficulty: A Level C2

The function f is defined by $f(x) = 2x^3 - x^2 - 4x - 4$.

Part A

Quotient

Find the quotient when $f(x)$ is divided by $x - 2$.

The following symbols may be useful: x

Part B

First stationary point of $f(x)$

Find the co-ordinates and nature of the stationary point of $f(x)$ with the larger x -coordinate.

The stationary point (,) is a .

Items:

-
-

Part C

Second stationary point of $f(x)$

Find the co-ordinates and nature of the stationary point of $f(x)$ with the smaller x -coordinate.

The stationary point (,) is a .

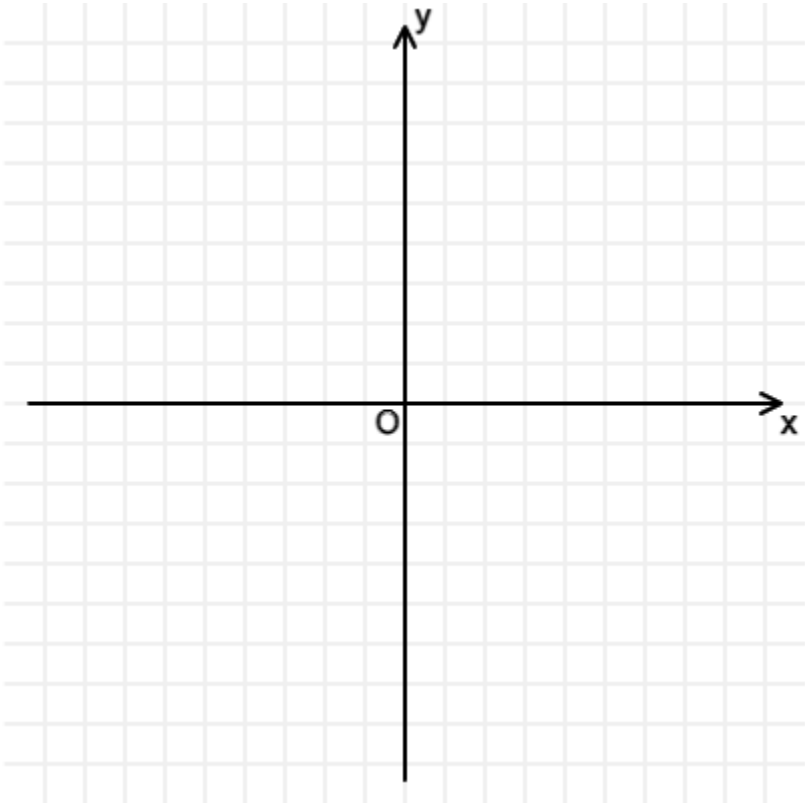
Items:

-
-

Part D

Sketch of $y = f(x)$

Sketch the graph of $y = f(x)$.



Part E

Sketch of $y = |f(x)|$

Sketch the graph of $y = |f(x)|$, then pick the graph that corresponds to $y = |f(x)|$ from the options below.

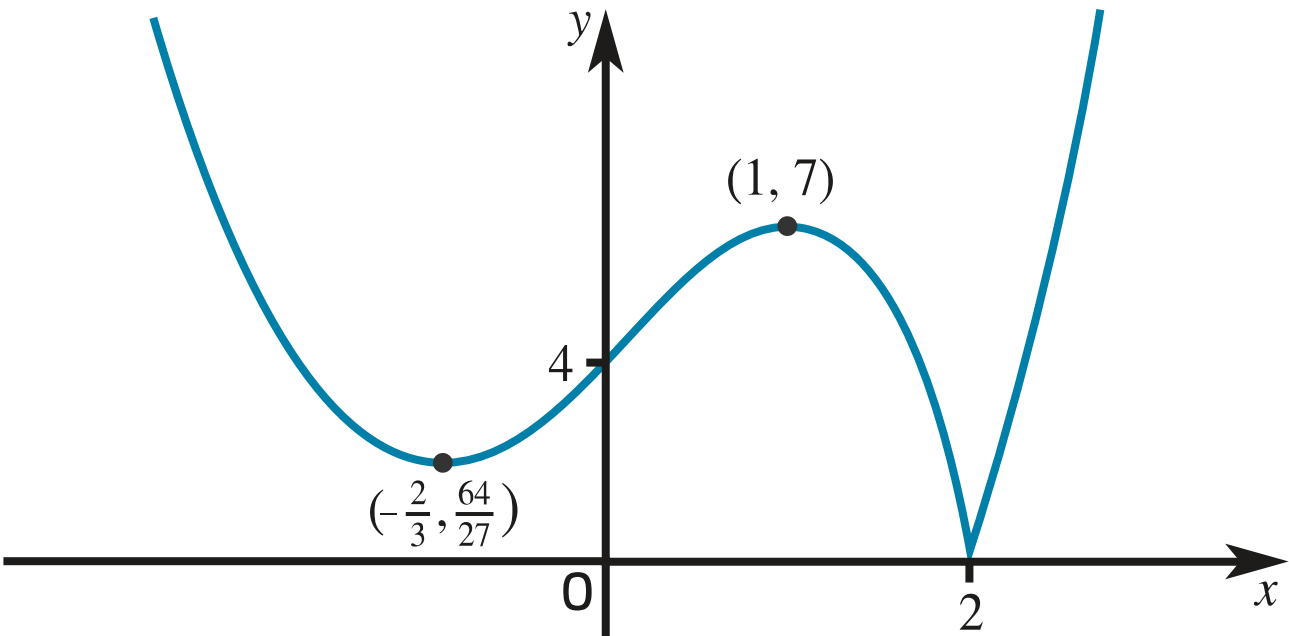


Figure 1: Option (i)

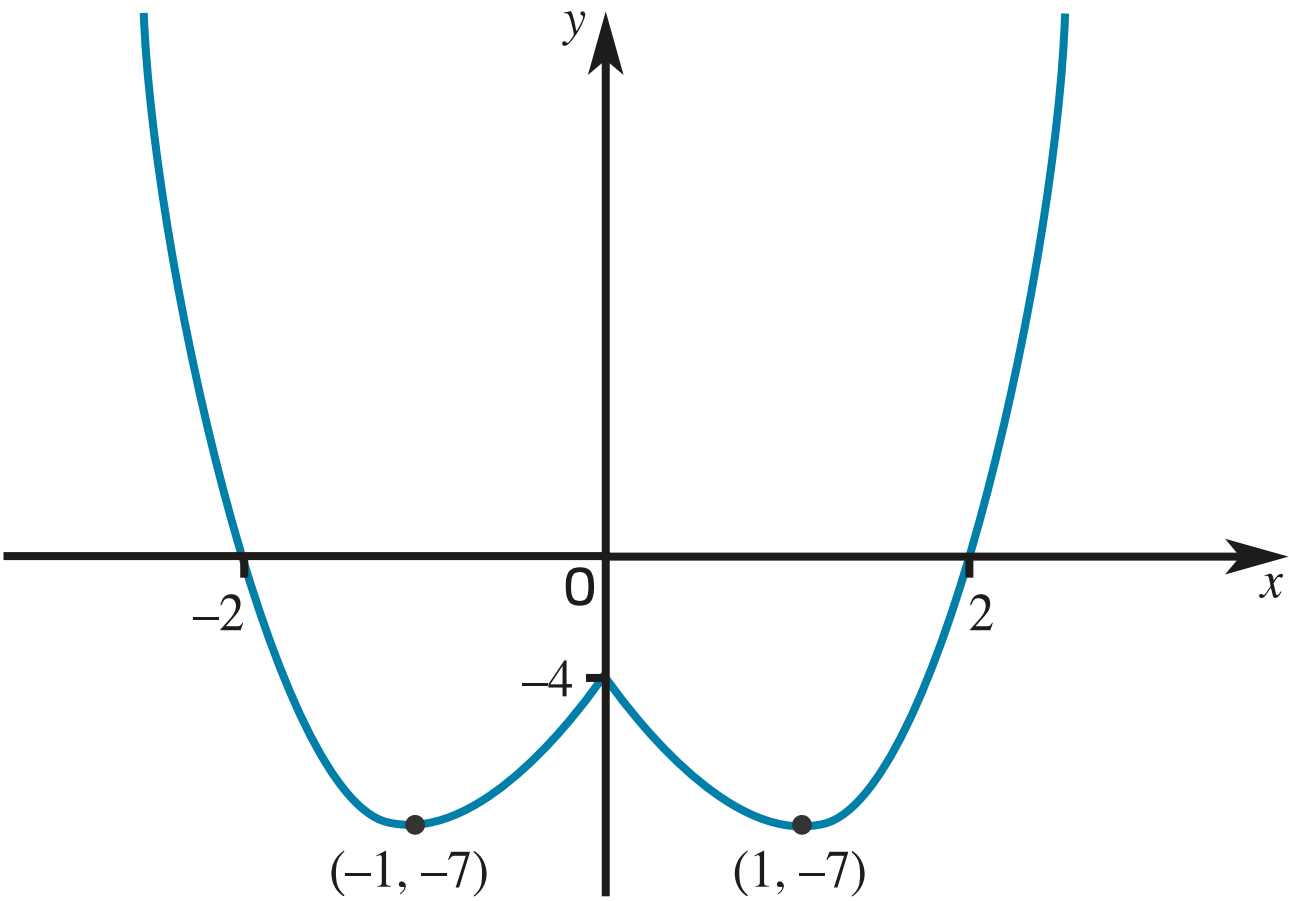


Figure 2: Option (ii)

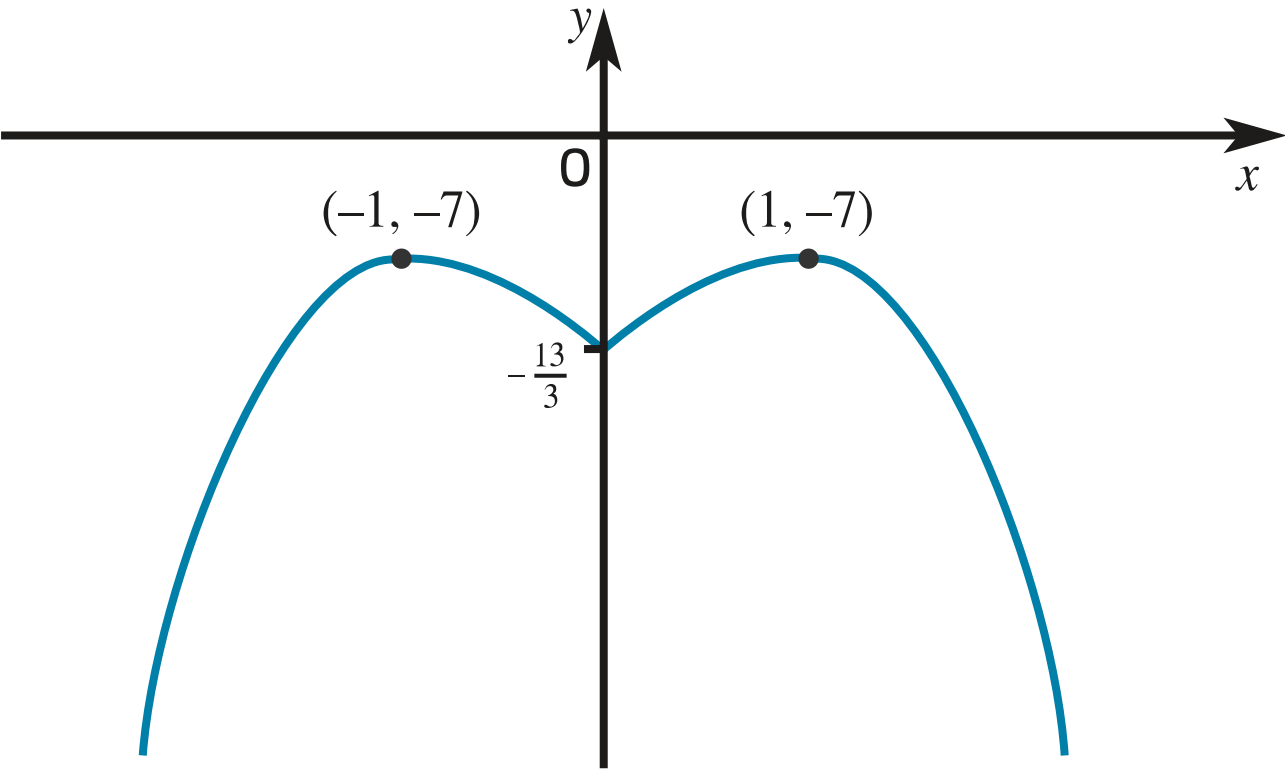


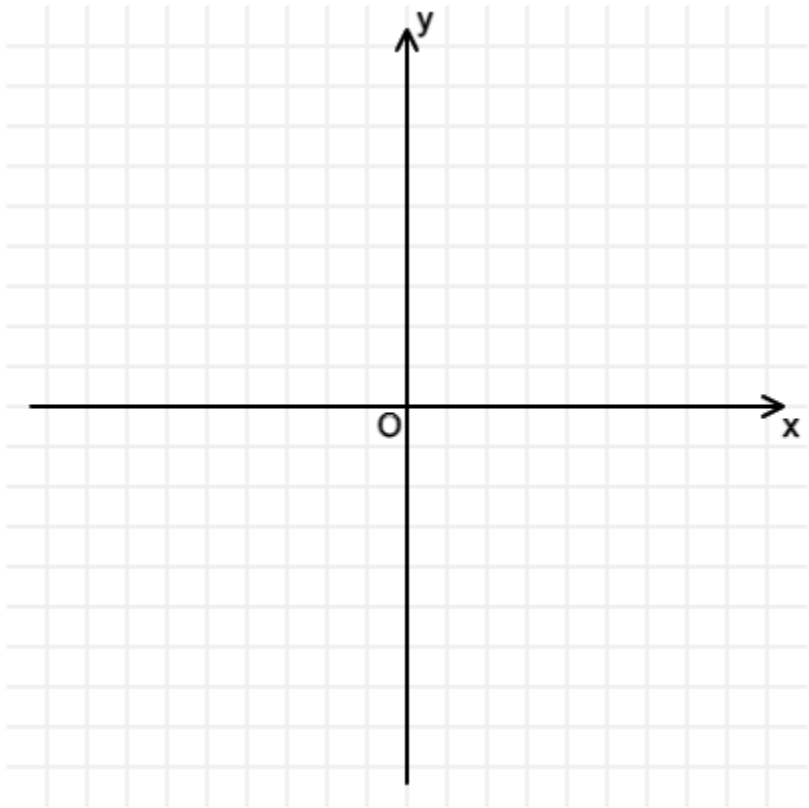
Figure 3: Option (iii)

- ☐ Option (i)
- ☐ Option (ii)
- ☐ Option (iii)

Part F

Sketch of $y = f(|x|)$

Sketch the graph of $y = f(|x|)$.
To prevent any sharp changes in your curve from being smoothed out, sketch your curve as two sections.



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Rational Inequality

Pre-Uni Maths for Sciences A2.10

Subject & topics: Maths | Functions | General Functions Stage & difficulty: Further A P2

Solve the inequality

$$\frac{x + 4}{x + 2} \leq \frac{x + 2}{x - 1}$$

giving your answer using set notation.

The solution is $\{x : \square\square x\square\square\} \cup \{x : x\square\square\}$.

Items:

-10

-9

-8

-7

-6

-5

-4

-3

-2

-1

0

1

2

3

4

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Rational Modulus Inequality

Pre-Uni Maths for Sciences E4.10

Subject & topics: Maths | Functions | General Functions Stage & difficulty: Further A P2

Solve the inequality

$$\frac{-x^2 - 5x + 24}{|x| + 3} > 2$$

giving your answer using set notation.

The solution is $\{x : \square\square x\square\square\}$.

Items:

−10

−9

−8

−7

−6

−5

−4

−3

−2

−1

0

1

2

3

4

5

6

7

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